

A Gold and Bitcoin Trading Model Based on XGBoost Prediction

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ABSTRACT

Transactions involving gold and Bitcoin have attracted a lot of attention. However, the existing trading institution models tend to focus on a single financial commodity and have certain limitations. This paper presents a new method for predicting gold and Bitcoin prices using XGBoost model. The performance of the model was evaluated using five predictors and the model prediction accuracy is high. In addition, taking into account principal, commission, risk and previous experience, the goal of this article is to establish the best one-day trading strategy for gold and Bitcoin. Using predictive price data, this paper uses genetic algorithms to optimize daily investment strategies and ultimately arrive at a plan with the highest potential return. This study provides valuable insights and recommendations for investors.

CCS CONCEPTS

• **Applied computing** → Operations research; Consumer products.

KEYWORDS

XGBoost, Best one-day trading strategy, Genetic algorithms

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1 INTRODUCTION

The trading of gold and Bitcoin has received much attention in recent years. Investors and traders constantly seek effective strategies to maximize returns in these markets. However, the existing models of trading institutions tend to focus on a single Financial instrument, such as gold or Bitcoin, while ignoring the potential advantages of both. This restriction can lead investors to make suboptimal trading decisions and miss opportunities. In the past, researchers such as Grudnitski and Osburn (1993) used neural networks to predict the S&P index and gold futures prices^[1], while Ismail, Yahya, and Shabri (2009) used multiple linear regression models to combine indicators such as commodities, exchange rates, and inflation rates to predict gold prices, with promising results^[2]. However, for ordinary individuals who lack professional knowledge

in various financial indicators, there is a need for a predictive model that is suitable for their needs. This study proposes a more easily obtainable method that utilizes only the historical prices of the first five days as input values, allowing for the development of more accurate models. The current research field lacks a comprehensive prediction and trading strategy development model that integrates gold and Bitcoin. Existing research mainly focuses on individual commodities and fails to capture the potential interrelationships and correlations between gold and Bitcoin prices. Therefore, a new approach is needed to consider these two types of assets and develop the best single day trading strategy.

2 PRICE PREDICTION MODEL BASED ON XGBOOST

2.1 XGBoost

XGBoost algorithm is an improvement of Boosting algorithm based on the tree model and GBDT(Gradient Boosting Decision Tree).The internal decision tree uses a regression tree, and the next tree is constructed according to the training residual of the previous tree, which is trained iteratively for many times, so as to establish a strong learner and gradually optimize the objective function to make the best prediction^[3].

To use the XGBoost algorithm to predict the price of gold and Bitcoin tomorrow, it needs to be trained in advance, and training requires a large amount of data, so we found the price data of gold and Bitcoin before 2016/9/12, and used the price data as the XGBoost learning data set^{[4][5][6][7]}.Taking into account the fluctuations of gold and Bitcoin prices, in the daily forecast, the price data of the first five days including the current day is regarded as reliable data, and it is used to predict the price of tomorrow. Due to the lack of data support on the first day, no prediction is made on the first day. The XGBoost algorithm is used to solve the price prediction problem^[8], and the price data of the previous five days is used as the characteristic data sample, so the data needs to be processed in steps, and the data is moved by 1-5 steps, representing the price data of the previous 1-5 days. Thereby, new five sets of data are generated, which are used as feature input. The process of price regression prediction using the XGBoost algorithm is shown in Figure 1.

2.2 Algorithm parameters tuning

Using grid search to find the optimal parameters, the parameters with the highest accuracy can be found within the specified parameter range, thereby reducing the complexity of the algorithm and improving the prediction accuracy of the model.The specific process is as follows:

1) According to the learning rate and the number of trees, specific parameters are tuned, including parameters such as the maximum

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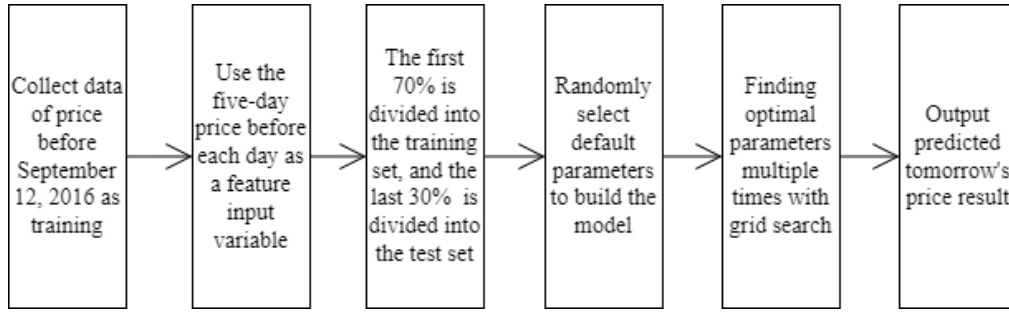


Figure 1: XGBoost operation flowchart.

Table 1: Parameters of the XGBoost model.

Parameter	Interpretation	Value (Gold)	Value (Bitcoin)
min_child_weight	The smallest leaf node: The smaller the weight, the easier it is to overfit.	3	3
learning_rate	Learning rate: Range 0 to 1, Commonly used 0.01~0.2, how many datasets are used for each iteration.	0.07	0.1
gamma	The declining value of the loss function: Sets the minimum threshold for loss function reduction, affecting node splitting	0.5	0.5
lambda	Regular term parameter: Defaults to 1, which controls the weight of the leaf node	1	1
max_depth	Maximum depth per tree: Common values 4 to 6, the deeper the depth, the easier it is to overfit	6	10

Table 2: Evaluation index values for training and test sets.

Gold	MSE	RMSE	MAE	MAPE	R^2
train set	64.943	8.059	5.543	0.516	1
test set	51.545	7.18	4.693	0.313	0.999
Bitcoin	MSE	RMSE	MAE	MAPE	R^2
train set	93.623	9.676	5.409	10.341	0.999
test set	18737.9	136.887	92.882	1.04	1

depth of the tree, the minimum leaf node weight, and the drop value of the loss function.

2) Tuning the regularization parameters to reduce the complexity of the model.

3) Adjust the learning rate within the default range to determine the optimal parameters. After several iterations of training, a set of optimal parameters is obtained, and the values are shown in Table 1.

2.3 Prediction results of price using XGBoost

The prediction results and real values of the regression model are continuous^{[9][10]}. The commonly used evaluation indicators of regression models are MSE, RMSE, MAPE, MAE, R^2 . The evaluation index values of the training set and test set are shown in Table 2

It can be seen that the accuracy of the XGBoost regression algorithm is high, and the average absolute percentage error can represent the error rate, which is only 1%; the degree of decision R^2 is close to 1, and the degree of compliance is extremely high. The training results can be used to predict gold and Bitcoin prices.

Using the model trained by XGBoost, the input is the price data of the previous five days, the output is the prediction of tomorrow's price, and the prediction result is run. Fit and compare the daily price forecasts of gold and Bitcoin with the actual value, as shown in Figures 2 and 3

3 ONE-DAY REVENUE PLANNING MODEL

3.1 Build mathematical model

Discussing single-day trading. Since we did not make predictions on the first day, we put 1/3 of our cash investment assets into the

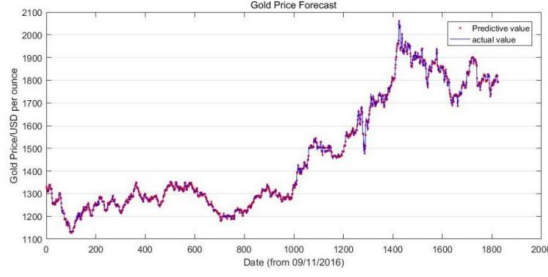


Figure 2: Comparison of Predicted Price and Actual Price (Gold).



Figure 3: Comparison of Predicted Price and Actual Price (Bitcoin).

gold market, 1/3 into the Bitcoin market, and the remaining 1/3 left. Traders before the day's trading The asset portfolio owned is {cash asset, gold asset, Bitcoin asset}, and its asset value is set as $\{C, G, B\}$, the trader needs to make a trading decision, and the daily trading volume of gold and Bitcoin is g respectively, b (when g, b is positive, it means buying gold or Bitcoin, and when it is negative, it means selling gold or Bitcoin, $a_{\text{gold}}=1\%$ and $a_{\text{Bitcoin}}=2\%$ are commissions for gold and Bitcoin), considering that the transaction requires transaction costs, the value of the asset portfolio after the transaction is:

$$C' = C - g - b - a_{\text{gold}} |g| - a_{\text{Bitcoin}} |b| \quad (1)$$

$$\begin{cases} G' = G + g \\ B' = B + b \end{cases} \quad (2)$$

Suppose today's gold and Bitcoin prices are n_1 and n_2 , respectively. Since the prices of tomorrow's gold and Bitcoin predicted by the model are used as a reference, and they are respectively set as m'_1, m'_2 , so the value of tomorrow's gold and Bitcoin assets is:

$$C'' = C' \quad (3)$$

$$\begin{cases} G'' = G' \frac{m'_1}{n_1} \\ B'' = B' \frac{m'_2}{n_2} \end{cases} \quad (4)$$

Let the predicted revenue generated by this transaction be v' , then:

$$\begin{aligned} v' &= (C'' + G'' + B'') - (C + G + B) \\ &= (G + g) \left(\frac{m'_1}{n_1} - 1 \right) + (B + b) \left(\frac{m'_2}{n_2} - 1 \right) \\ &\quad - a_{\text{gold}} |g| - a_{\text{Bitcoin}} |b| \end{aligned} \quad (5)$$

In actual investment, traders will not only consider the benefits, but also the size of the risk, and the traders will consider the error between the predicted result and the actual price, so the risk is introduced into the model. Risk is a non-quantitative concept, so the model first needs to quantify risk. In practice, the risk is related to the fluctuation of the currency price, so the model takes the price fluctuation range of the previous week (the first seven days are the first day to the current day) as the risk quantitative indicator, and the risk coefficients of gold and Bitcoin transactions are set as r_1, r_2 , we define:

$$r_{1,2} = \frac{1}{7} \sum_i d \quad (6)$$

$$d_{1,2} = \left| \frac{m_{1,2}}{n_{1,2}} - 1 \right| \quad (7)$$

At the same time, the transaction risk is related to the transaction amount. The larger the transaction amount, the greater the risk borne. Therefore, the risk amount is set as q , which is defined as follows:

$$q = d_1 |g| + d_2 |b| \quad (8)$$

In the model, it is assumed that traders regard risk and return as equally important, that is, assign a weight value $w=0.5$ to each risk, and establish a comprehensive evaluation function of risk and return:

$$S = (1 - w) v' - w q \quad (9)$$

The evaluation result S is regarded as the decision-making target, and the evaluation value is positively correlated with the benefit and negatively correlated with the risk, so the higher the evaluation value, the better the decision-making.

Turning the problem into an optimization problem, the model needs to solve the maximum value of the daily transaction evaluation value S , and the decision variable is the daily gold and Bitcoin transaction volume. Consider the constraints:

1. The transaction amount and transaction cost shall not exceed the principal amount;
 2. The quantity of gold and Bitcoin sold cannot exceed the holding amount;
 3. Gold cannot be traded on non-trading days.
- So the planning model is obtained as follows:

$$\begin{aligned} &\text{Min } wq - (1 - w) v' \\ &\text{s.t. } \begin{cases} (1 + \alpha_{\text{gold}}) \times g + (1 + \alpha_{\text{Bitcoin}}) \times b \leq C \\ -G \leq g \\ -B \leq b \end{cases} \end{aligned} \quad (10)$$

On non-gold trading days, the constraints are as follows:

$$\begin{aligned} &\text{s.t. } \begin{cases} (1 + \alpha_{\text{gold}}) \times g + (1 + \alpha_{\text{Bitcoin}}) \times b \leq C \\ g = 0 \\ -B \leq b \end{cases} \end{aligned} \quad (11)$$

3.2 Genetic Algorithm

According to the concept of biological evolvability, the pursuit of genetic algorithms is also the ability of the current population to produce better individuals than the existing individuals, that is, the evolutionary ability of genetic algorithms or population evolvability. Therefore, the research of genetic algorithm theory and method revolves around this goal. Genetic algorithm simulates reproduction, crossover and gene mutation in the process of natural selection and natural heredity, retains a set of candidate solutions in each iteration, selects the best individuals from the solution group according to some index, combines these individuals by using genetic operators (selection, crossover and variation) to produce a new generation of candidate solutions, and repeats this process until a certain convergence index is met. In variable processing, it is necessary to carry out coding operation. Coding is a method to convert the feasible solution of a problem from its solution space to the search space that genetic algorithm can handle. In the problem of genetic algorithm coding, the variable is transformed into binary coding chromosome in this paper. The operator of genetic algorithm includes three basic forms: selection, crossover and mutation, which constitute the core of the powerful search ability of genetic algorithm, and are the main carriers to simulate reproduction, hybridization and mutation in the process of natural selection and heredity^{[12][13]}. In the selection operation, this paper chooses the traditional roulette mode^[14]. The roulette selection strategy is to calculate the probability of each individual appearing in the offspring according to the fitness value of the individual, and select the individual randomly according to this probability to form the offspring population^[15]. The starting point of roulette selection strategy is that the better the fitness value, the greater the probability of being selected. The calculation is small and the effect is good.

3.3 Model solving

The utilization of Genetic Algorithm (GA) for optimizing trading volume exhibits several advantageous traits. GA possesses the capability to optimize the search range effectively, boasting robust global search ability while operating with minimal constraints and rules. By leveraging GA to determine the optimal value of the calculation function, one can establish a fitness function, subsequently allowing for the automatic minimization of this function. This approach facilitates rapid convergence of calculations, thereby demonstrating robust performance under varying conditions^{[16][17]}. In the context of optimizing trading strategies, GA is adeptly employed to refine decision-making processes. To ensure the preservation of genes associated with the most optimal strategies, it becomes imperative to assign higher fitness to decisions yielding superior evaluation values. As such, a fitness function, denoted as $g(x)$, is meticulously defined. This function plays a pivotal role in prioritizing decisions based on their performance, thereby enhancing the overall efficacy of the trading strategy optimization process.

$$g(x) = \frac{1}{f(x) - f(x)_{\min} + \epsilon} = \frac{1}{-S + S_{\max} + \epsilon} \quad (12)$$

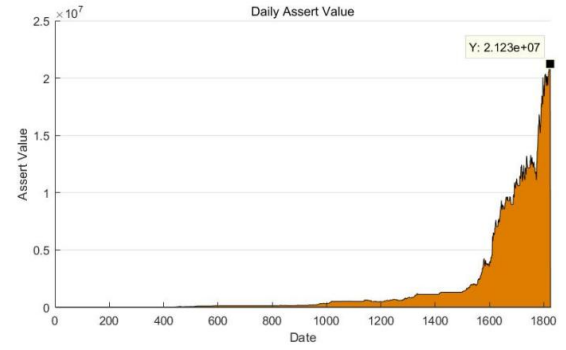


Figure 4: Daily asset trend.

Set the population size to 1000, the probability of crossover and mutation to 0.6 and 0.01, respectively, and the number of evolutions to 10 to find the best daily trading strategy (g_{best} , b_{best}). Invest according to the best daily trading strategy, and the actual assets after daily trading to the next day are:

$$\begin{cases} C_{i+1} = C_i - g_{best} - b_{best} - a_{gold} |g_{best}| - a_{Bitcoin} |b_{best}| \\ G_{i+1} = (G_i + g_{best}) \frac{m_1}{n_1} \\ B_{i+1} = (B_i + b_{best}) \frac{m_2}{n_2} \end{cases} \quad (13)$$

Under this plan, the initial principal is \$1,000, and the optimal strategic plan is invested every day, and then optimized the next day after the investment. Repeat this process to get the total assets for each day, as shown in the Figure 4

4 CONCLUSION

This article employs the XGBoost prediction model, utilizing historical prices as state variables, to construct a simple and accurate prediction model. Leveraging the forecasted data, it establishes a risk-sensitive investment strategy for gold and Bitcoin with the aim of maximizing daily profits. In contrast to traditional algorithms, the article utilizes a genetic algorithm for optimization problem-solving, resulting in faster convergence. The genetic algorithm relies solely on the value of the objective function as search information, thereby reducing the complexity of the solution. However, there are also drawbacks. For example, a single genetic algorithm may not comprehensively impose constraints, necessitating an increase in decision variable thresholds, which in turn leads to a higher workload and longer solving time.

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